Final

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Set Up

The link to the GitHub repo can be found here.

```
# read in packages
library(tidyverse)
library(here)
library(DHARMa)
library(MuMIn)
library(ggeffects)
library(janitor)

# read in data
nest <- read_csv(here("data", "nest_data_final.csv"))
walk <- read_csv(here("data", "ben-data-tracker.csv"))
```

Problem 1

a.

In part 1, my coworker used a `generalized linear model`. I can tell because the results state a probability between two binary outcomes, the American Avocet using the habit or the American Avocet not using the habitat. This response variable is binary, which necessitates a generalized linear model to look at the relationship between that and the predictor of distance to the water.

In part 2, my coworker used a `chi-squared test`. I can tell because the results compare two categorical variables: habitat and bird species. Looking at the relationship between two categorical variables and seeing if one has an influence on the other requires using a chi-squared test.

b.

For part 1: We found that distance to the water's edge significantly influences the probability of birds utilizing microhabitats.

For part 2: We found a significant relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and the habitats they are found in (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat). Different aquatic bird species generally occupy different aquatic habitats.

c.

Water depth may influence the probability of microhabitat use. The microhabitats are all aquatic, and water depth may influence things like the ability for birds to wade and the availability of various food sources, which, in turn, would affect which bird species use the microhabitats and how they use it. Water depth is a continuous variable which can be measured in a unit such as cm or meters.

Problem 2

a.

```
# create new object nest_clean from nest
nest_clean <- nest |>
  # only keep the columns for ant species, tree height and nest presence
  # (case_control)
  select(ant_species, height_cm, case_control) |>
  # change species to full Latin names
  mutate(ant_species = case_when(
    ant_species == "RRB" ~ "C. mimosae",
    ant_species == "BBR" ~ "C. nigriceps",
    ant_species == "AB" ~ "C. sjostedti"
  )) |>
  # remove NA values to only keep species we're interested in
  drop_na() |>
  # assign levels to each ant species for later analysis
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "C. sjostedti",
                                   "C. mimosae",
                                   "C. nigriceps"
                                   ))
# display the structure of the cleaned data (confirms species has levels)
str(nest_clean)
```

```
tibble [270 x 3] (S3: tbl_df/tbl/data.frame)
 $ ant_species : Factor w/ 3 levels "C. sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
```

```r
# view 10 random rows from the cleaned data frame
slice_sample(
  nest_clean,
  n = 10
)
```

```
# A tibble: 10 x 3
   ant_species  height_cm case_control
   <fct>            <dbl>        <dbl>
 1 C. mimosae       332.             0
 2 C. mimosae       199              0
 3 C. nigriceps      65              0
 4 C. mimosae       432.             0
 5 C. nigriceps     163              0
 6 C. mimosae       116.             0
 7 C. sjostedti     156              0
 8 C. nigriceps      96.5            0
 9 C. mimosae       125              0
10 C. nigriceps     246              1
```

b.

The 1s in the dataset represent trees that have bird nests in them, and the 0s represent trees that do not have bird nests in them (called "available" in the study).

c.

Tree Height

Tree height is a continuous variable. The study used two logistic models, one of which predicted that height positively influences nest probability (Results, paragraph 2). However, the other model predicted that height has no influence on nest probability (Results, paragraph 3).

Acacia Ant Species

Ant species is a categorical variable with 3 levels. The study stated that *C. mimosae* and *C. nigriceps* are aggressive species while *C. sjoestedti* is not aggressive (Introduction, paragraph 2). The study predicted that aggressive ants may offer protection to bird nests, but it notes that aggressive ants may also attack bird nests (Introduction, paragraph 3). The models generated predicted that trees with *C. nigriceps* had a greater probability of nests than trees with *C. mimosae* (Results, paragraph 2) and that trees with *C. mimosae* had a greater probability of nests than trees with *C. sjostedti* (Results, paragraph 3).

Neither model in the study used an interaction term (Methods, paragraph 2).

d.

$+$ indicates no interaction, X indicates an interaction

Model number	Height	Ant Species	Model Description
0	+	+	Both terms, no interaction
1	X	X	Both terms, interaction term

e.

```r
# model 0: glm with no interaction term
no_int <- glm(
  # formula for all predictors, no interaction
  case_control ~ height_cm + ant_species,
  # use cleaned data frame for nest data
  data = nest_clean,
  # set family to binomial due to the binomial response variable
  family = "binomial"
)

# model 1: glm with ant species as an interaction term with height
int_term <- glm(
  # asterisk to indicate an interaction
  case_control ~ height_cm * ant_species,
  # use cleaned data
  data = nest_clean,
  # set family to binomial due to the binomial response variable
```

```
  family = "binomial"
)
```

f.

```
# put both models into calculating AICc
AICc(
  no_int,
  int_term
) |>
  # sorts the AICc values so the lowest (best model) appears first
  arrange(AICc)
```

```
         df     AICc
int_term  6 238.1236
no_int    4 258.8940
```

```
# provide summary statistics for the interaction term model
summary(int_term)
```

```

Call:
glm(formula = case_control ~ height_cm * ant_species, family = "binomial",
    data = nest_clean)

Coefficients:
                                   Estimate Std. Error z value Pr(>|z|)
(Intercept)                       -3.693410   2.517395  -1.467  0.14233
height_cm                          0.003895   0.008237   0.473  0.63634
ant_speciesC. mimosae              0.848334   2.554617   0.332  0.73983
ant_speciesC. nigriceps          -12.279738   5.946723  -2.065  0.03893 *
height_cm:ant_speciesC. mimosae    0.002597   0.008386   0.310  0.75679
height_cm:ant_speciesC. nigriceps  0.084541   0.031961   2.645  0.00817 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 286.04  on 269  degrees of freedom
```

```
Residual deviance: 225.80  on 264  degrees of freedom
AIC: 237.8

Number of Fisher Scoring iterations: 7
```

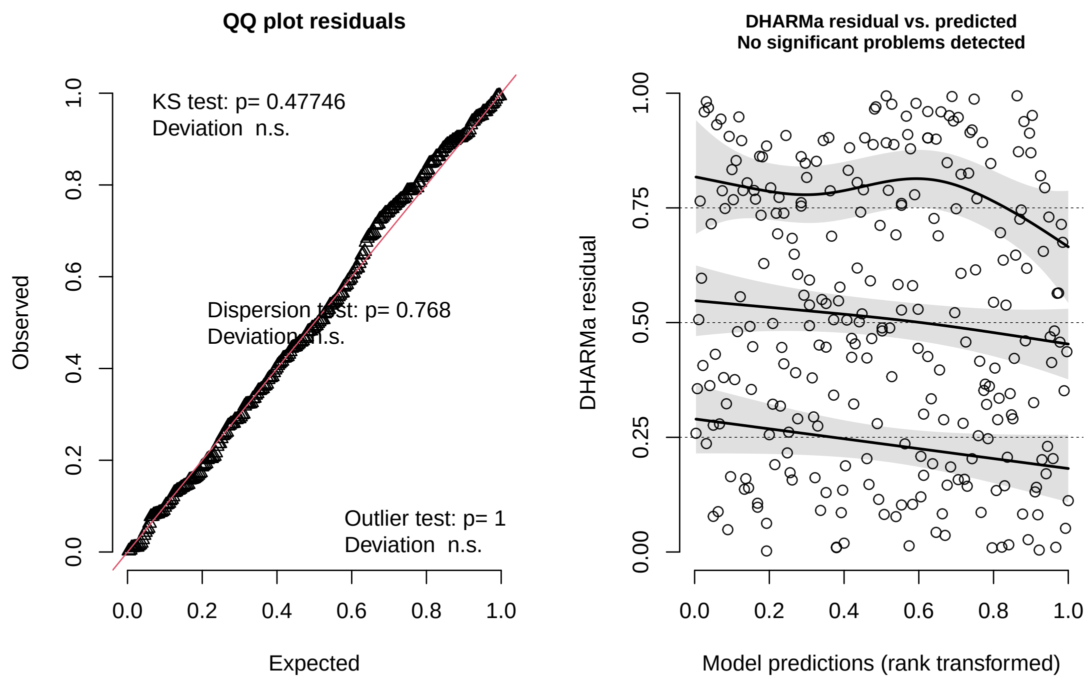
The AICc for the model with an interaction term is 238.12 while the one without an interaction term is 258.89. The model with an interaction term is significantly higher (that is, it exceeds the other model by more than 2), so this is the best model for the system.

The best model as determined by Akaike's Information Criterion (AIC) was the model where Acacia Ant species and tree height interact to determine the probability of a bird nest being present in the tree. The effect of tree height on the probability of a nest varies based on the ant species present in the tree.

g.

```r
# display diagnostics for the glm with the interaction term
plot(
  simulateResiduals(int_term)
  )
```

DHARMa residual



The QQ plot residuals roughly follow the reference line for a uniform distribution. The residuals vs predicted plot appears random with no noticeable patterns. From this, both diagnostic plots show that the glm assumptions are met.

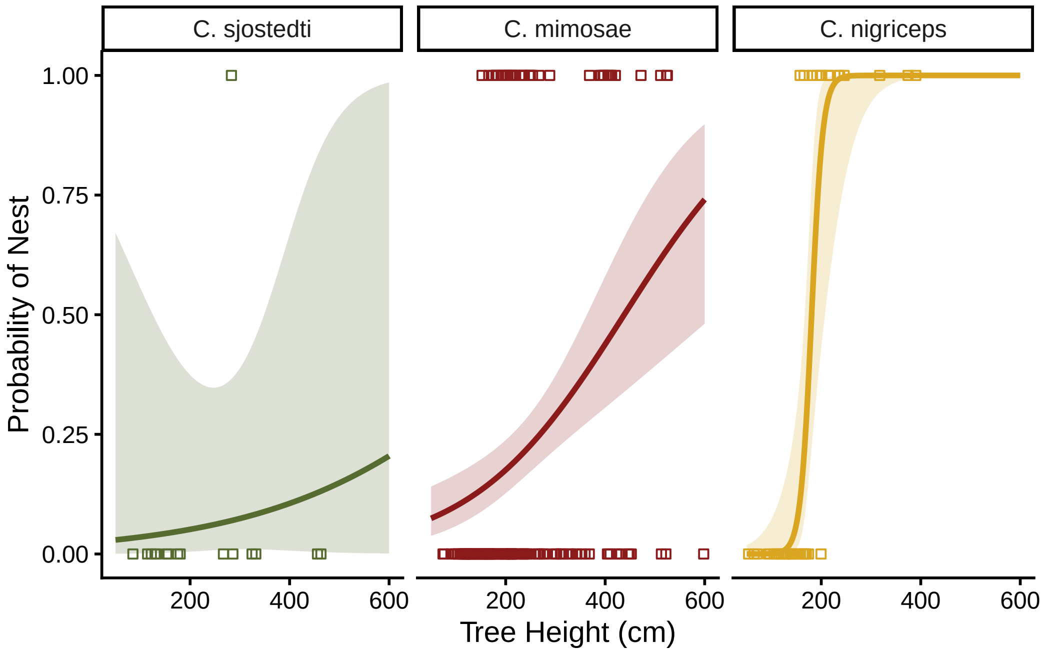
h.

```
# create a new data frame that stores predictions from the model
nest_predict <- ggpredict(
  # use the model with the interaction term
  int_term,
  # create predictions for nest probability from height by every cm
  # between 50-600cm, also consider ant species
  terms = c(
    "height_cm [50:600 by = 1]",
    "ant_species")
) |>
  # rename model data frame columns to align with the nest_clean data frame
  rename(
```

```
    height_cm = x,
    case_control = predicted,
    ant_species = group
  )
```

```
# create the base layer using the nest_clean data frame
ggplot(
  data = nest_clean,
  # continuous predictor, height_cm, on x-axis
  # binomial response, case_control (nest probability), on y-axis
  aes(
    x = height_cm,
    y = case_control
  )
) +
  # layer 1: create points for the underlying data
  geom_point(
    # color and group data by ant species
    aes(color = ant_species),
    # using an outlined shape to make data distinguishable and to ensure the
    # model line stands out more
    shape = 22
  ) +
  # layer 2: create the model line for each ant species
  geom_line(
    # switch to using the model predictions data
    data = nest_predict,
    # use the same axes and groupings as before
    aes(
      x = height_cm,
      y = case_control,
      color = ant_species
    ),
    # ensure line is thick enough to be seen clearly
    linewidth = 1
  ) +
  # layer 3: 95% confidence interval for nest probability, as a ribbon
  geom_ribbon(
    # continue using the nest_predict data frame
    data = nest_predict,
    # 95% CI is pulled from conf.low and conf.high, use fill for color
    aes(
```

```
      x = height_cm,
      y = case_control,
      ymin = conf.low,
      ymax = conf.high,
      fill = ant_species
    ),
    # make ribbon transparent to ensure line stands out
    alpha = 0.2
  ) +
  # create a different plot for each ant species
  facet_wrap(~ ant_species) +
  # adjust line and point colors
  scale_color_manual(values = c(
    "C. sjostedti" = "darkolivegreen",
    "C. mimosae" = "firebrick4",
    "C. nigriceps" = "goldenrod"
  )) +
  # adjust ribbon color
  scale_fill_manual(values = c(
    "C. sjostedti" = "darkolivegreen",
    "C. mimosae" = "firebrick4",
    "C. nigriceps" = "goldenrod"
  )) +
  # adjust x-axis to only include heights within nest_clean
  scale_x_continuous(
    limits = c(50, 600)
  ) +
  # rename axes to be more descriptive
  labs(
    x = "Tree Height (cm)",
    y = "Probability of Nest"
  ) +
  # remove gridlines and the grey background
  theme_classic() +
  # remove the redundant legend
  theme(legend.position = "none")
```



i.

Figure 1

Tree Height and Acacia Ant Species Influences the Probability of Bird Nesting

Colored lines represent the predicted probability of finding a bird nest in a tree as a function of tree height (in cm) and ant species (*Creanomaster sjostedti*, *C. mimosae* and *C. nigriceps*) present in the tree. Probabilities were calculated using a generalized linear model (GLM). The shaded ribbons around the line represent the 95% confidence interval for the probability of finding a bird nest. The outlined squares represent the underlying data. The different colors and plots represent the data and model for an individual ant species. Data originally from: Lujan. E, et al (2023). Symbiotic acacia ants drive nesting behavior by birds in an African savanna. Biotropica, Volume 55, Issue 6. https://doi.org/10.1111/btp.13276. Accessed 2026/06/04.

j.

```
# use the interaction term model to create a prediction at height 600cm
ggpredict(
  # use the interaction term model
```

```
  int_term,
  # make the prediction at height 600cm for all ant species
  terms =
    c("height_cm [600]", "ant_species")
)
```

```
# Predicted probabilities of case_control

ant_species: C. sjostedti

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

ant_species: C. mimosae

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

ant_species: C. nigriceps

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k.

We found that the height of an acacia tree and the species of Acacia Ant (*Creanomaster sjostedti*, *C. mimosae* and *C. nigriceps*) present influences the probability of finding a bird nest in the tree. All trees had a greater probability of having nests as height increased, but the probability of nests increased as a function of height more in trees with *C. mimosae* than trees with *C. sjostedti* and even more so in trees with *C. nigriceps* than for trees with either of the other species. At a tree height of 600cm, trees with *C. sjostedti* had a 20% chance (95% CI: [0.00, 0.99]) of a nest, trees with *C. mimosae* had a 74% chance (95% CI: [0.48, 0.90]) of a nest and trees with *C. nigriceps* had a near 100% chance (95% CI: [1.00, 1.00]) of a nest. *C. mimosae* and *C. nigriceps* are considered aggressive ant species while *C. sjoestedti* is not aggressive. Despite this aggression, the ants do not attack bird nests; rather, the birds benefit from ant aggression since it drives away nest predators, explaining why bird nests are more likely to be found in trees with aggressive ant species.

Problem 3

a.

My visualizations from homework 2 were barebones, straightforward and only analyze one predictor variable at a time (lunch location and temperature in relation to distance walked), and it's also clear what the predictor is versus the response (based on axes). By contrast, my affective visualization is abstract, looks at many predictors (day of week, calendar events and bus trips in relation to distance walked) and does not clearly distinguish what the response variable is; everything interacts with each other. Neither predictor I used in my homework 2 visualizations were used in the affective visualization.

However, all visualizations include the response variable, distance walked, in them.

My homework 2 visualizations don't really show any apparent relationship between the predictor and response (there may be a slight correlation for temperature). It's much harder to piece together an idea of relationships from the affective visualization, but it appears that calendar events and day of the week had an influence on distance walked (this was more apparent in my infographic than the affective visualization).

My feedback from workshop 9 was to make the visualization larger so that the small details, which represent the data, could be seen, which is why I used multiple pages in my final piece (it also gave it the feel of flipping through a calendar, which I intended for). I also changed how calendar events was represented to make it easier to read, and I decided to show day of the week rather than temperature since I thought that would be more interesting.

b.

Temperature got no mention in my infographic or affective visualization, so I think it's time to give it some love, and I want to know if there was a statistically significant relationship. Although my primary question was seeing if day of the week influenced distance walked, I also wanted to know how my other predictors influenced it, and one of those predictors was temperature. I predicted that as temperature went up, my distance walked would decrease since I wouldn't want to walk in the heat. The best way to see if this relationship exists is to use a `linear model`.

Temperature is a continuous variable. My app for measuring temperature was only accurate to the nearest degree, which is why all data took on integer values. However, a prediction for a temperature of 17.5° C makes sense (as opposed to a prediction for a discrete variable, such as a count of 17.5 elephants).

Distance walked is also a continuous variable that I was able to measure down to a hundredth of a kilometer.

c.

Assumptions & Analysis

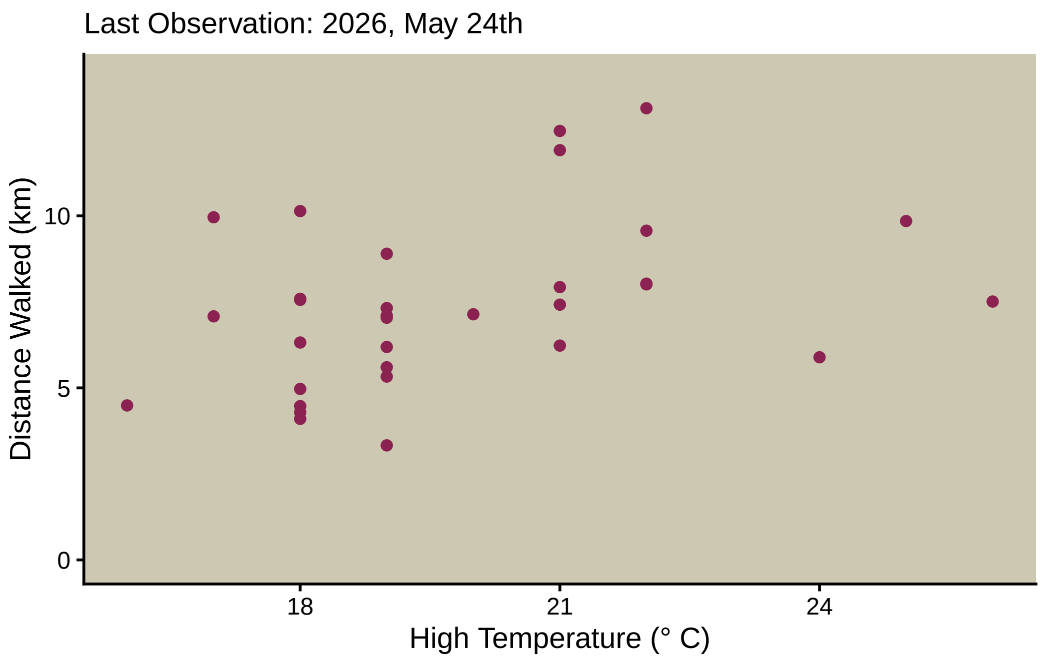
Before fitting the model, we need to need to confirm that errors are independent. Based on how I collected data, each observation is independent, so errors should be independent too. We also need to confirm linearity.

Here is an exact copy of the code/visualization from homework 3 (with all of the data).

```r
# create a new object to clean data
walk_clean <- walk |>
  # remove uppercases and add underscores
  clean_names() |>
  # remove N/A values
  drop_na()

# create a scatterplot for the data using walk_clean
ggplot(walk_clean,
       # x-axis: high temperature (in celsius)
       # y-axis: distance walked (in km)
       # color: set color by lunch location
       mapping = aes(x = high_temperature_c,
                     y = distance_walked_km)) +
  # once again starting y from 0
  # It would not make sense to set temperature to 0 too
  # (this would make it hard to visualize the data's spread),
  # so I am leaving the x-axis as is
  scale_y_continuous(limits=c(0, 14)) +
  # create a scatterplot with pink points
  geom_point(color = "violetred4") +
  # setting theme to classic
  theme_classic() +
  # removing legend
  theme(legend.position = "None") +
  # renaming axes and title
  labs(
    # renaming x-axis
    x = "High Temperature (\U00B0 C)",
    # renaming y-axis
    y = "Distance Walked (km)",
    # creating subtitle with last observation
    subtitle = "Last Observation: 2026, May 24th"
```

```
  ) +
  # changing colors of various elements to improve contrast
  theme(
    # change x-axis color
    axis.text.x = element_text(color = "black"),
    # change y-axis color
    axis.text.y = element_text(color = "black"),
    # change panel background
    panel.background = element_rect(fill = "cornsilk3"),
    # change panel line colors to remove them
    panel.grid = element_line(color = "cornsilk3"))
```



Although the data is scattered, it appears there could be some sort of linear relationship (a slightly positive relationship). Furthermore, it doesn't appear like the data follows a relationship that is not linear; it's either linear or there's no relationship. It would also make sense for this relationship to be linear, especailly over such a small range of temperatures.

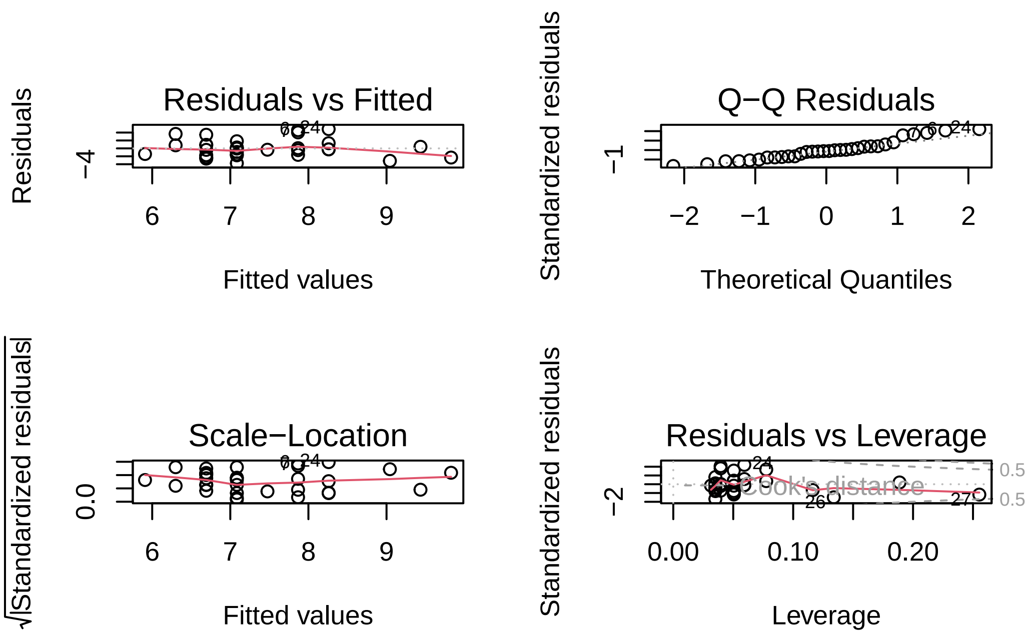
We now need to run the model to check for homoscedasticity and normal distribution of the residuals

```
# create a linear model of distance walked as a function of temperature
# use the cleaned walking data frame
walk_model <- lm(
```

```
  distance_walked_km ~ high_temperature_c,
  data = walk_clean
  )
```

Now we can check for the other two assumptions using the diagnostics plots.

```
# displays plots in a 2x2 grid using the linear model
par(mfrow = c(2, 2))
plot(walk_model)
```



The Residuals vs Fitted plot and the Scale-Location plot have no apparent patterns, so the residuals are homoscedastic. The QQ plot of the residuals shows a deviation in the upper tail, but the center falls on the reference line for normality, and the deviation is not severe, so the residuals are normal enough to meet assumptions. Furthermore, the Residuals vs Leverage plot shows no outliers in the data since all data is inside Cook's Distance.

Now we can get the summary statistics and a table of predictions for the model.

```
# summarize model to get summary statistics
summary(walk_model)
```

```
Call:
lm(formula = distance_walked_km ~ high_temperature_c, data = walk_clean)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.7538 -1.6586 -0.2393  0.8755  4.8707 

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)  
(Intercept)         -0.3608     3.4693  -0.104   0.9179  
high_temperature_c   0.3918     0.1739   2.253   0.0317 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.274 on 30 degrees of freedom
Multiple R-squared:  0.1447,    Adjusted R-squared:  0.1162 
F-statistic: 5.075 on 1 and 30 DF,  p-value: 0.03174
```

```r
# generate a data frame of predictions from the model based on the
# high temperature (will be used in visualization)
walk_predict <- ggpredict(
  walk_model,
  terms = "high_temperature_c"
) |>
  # rename columns to align with walk_clean data frame
  rename(
    high_temperature_c = x,
    distance_walked_km = predicted
  )
```

The summary shows that the residual median is a bit off from 0 (-0.239), but not so far that it suggests non-normality in the residuals. The quartiles and min/max also don't line up perfectly, but once again, not so much that it suggests non-normality.

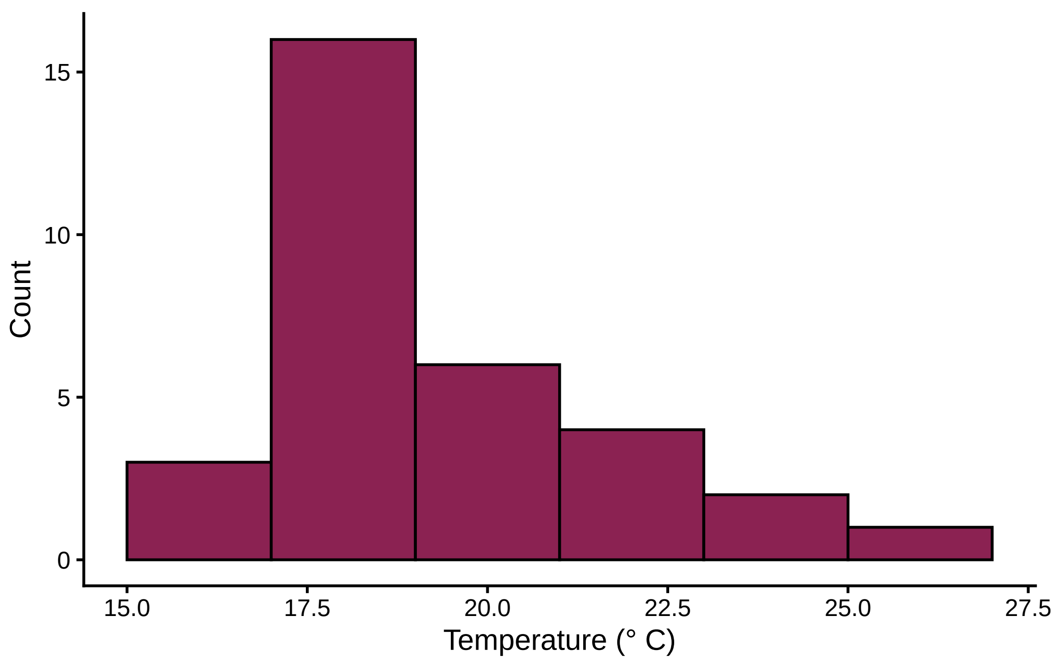
As for the summary statistics, the equation ends up being: `distance walked = -0.36 + 0.3918 * temperature`

Note that the intercept is not statistically significant (p = 0.9197), but the slope is (p = 0.0317) using a significance level of $\alpha = 0.05$.

Correlation (Effect Size)

We can run Pearson's Correlation to see how strong of a relationship there is. This requires checking that both temperature and distance walked are normally distributed.

```
# create a histogram for the temperature
# use the cleaned walking data and use the temperature column
ggplot(
  data = walk_clean,
  aes(x = high_temperature_c)) +
  # create histogram with custom color, Rice Rule says to use 6 bins
  geom_histogram(bins = 6,
                 color = "black",
                 fill = "violetred4") +
  # rename axes
  labs(
    x = "Temperature (\U00B0 C)",
    y = "Count"
  ) +
  # change theme to remove gridlines
  theme_classic()
```



There is enough right skewness here to violate the assumption of normality. However, we can run Spearman's correlation instead. This requires that the data is monotonic. Based on the initial visualization, it doesn't appear like the data is going down and then up or vice versa; it's either consistently trending in one direction or not trending in any direction. Therefore, we can run Spearman's correlation.

```
# run spearman's correlation test
cor.test(walk_clean$high_temperature_c, walk_clean$distance_walked_km,
         method = "spearman",
         exact = FALSE)
```

```

	Spearman's rank correlation rho

data:  walk_clean$high_temperature_c and walk_clean$distance_walked_km
S = 3190.3, p-value = 0.0181
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho 
0.4152681 
```

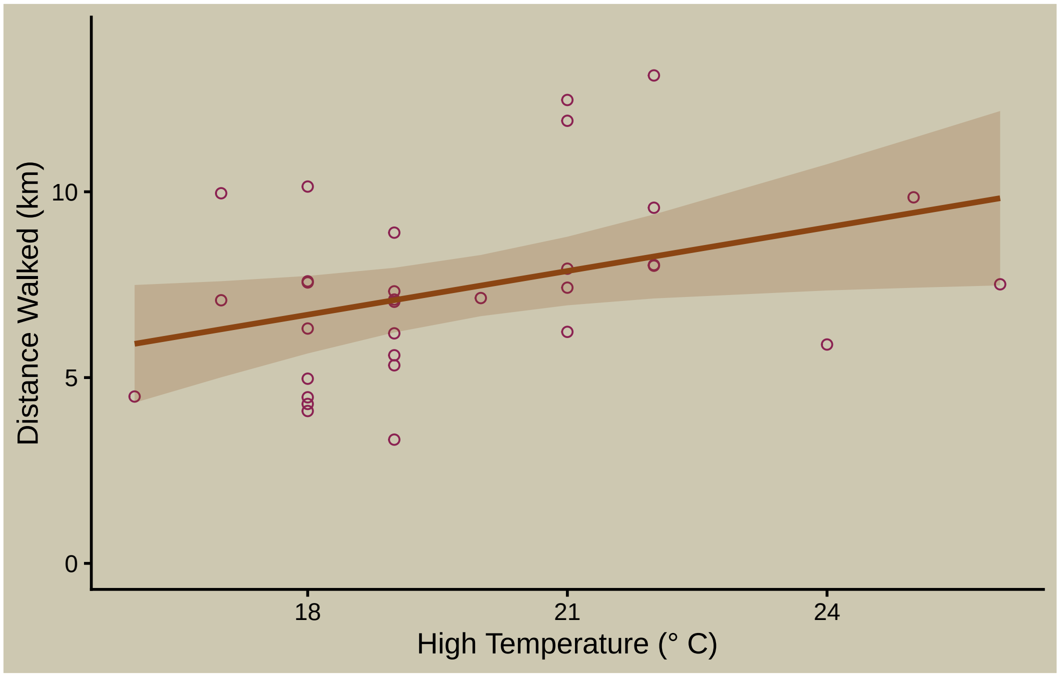
The test returns $\rho = 0.415$ (moderate correlation) and p = 0.018

d.

We can overlay the original visualization with the model.

```
# layer 1: create a scatterplot for the underlying data
ggplot(walk_clean,
       mapping = aes(x = high_temperature_c,
                     y = distance_walked_km)) +
  # scale y to start at 0
  scale_y_continuous(limits=c(0, 14)) +
  # create a scatterplot with pink points, make points large but outlined
  geom_point(color = "violetred4",
             size = 1.5,
             shape = 21) +
  # layer 2: show the 95% confidence interval
  # use ribbon gemoetry using the walking model predictions
  geom_ribbon(data = walk_predict,
              aes(x = high_temperature_c,
                  y = distance_walked_km,
                  ymin = conf.low,
```

```
                   ymax = conf.high),
               # make ribbon different color than points
               fill = "chocolate4",
               # make ribbon transparent
               alpha = 0.2) +
  geom_line(data = walk_predict,
               aes(x = high_temperature_c,
                   y = distance_walked_km,
                   ),
               # have line color match ribbon color
               color = "chocolate4",
               # make line thick so it's visible
               linewidth = 1
             ) +
  # setting theme to classic
  theme_classic() +
  # removing legend
  theme(legend.position = "None") +
  # renaming axes and title
  labs(
    x = "High Temperature (\U00B0 C)",
    y = "Distance Walked (km)"
  ) +
  # changing colors of various elements
  theme(
    axis.text.x = element_text(color = "black"),
    axis.text.y = element_text(color = "black"),
    panel.background = element_rect(fill = "cornsilk3"),
    panel.grid = element_line(color = "cornsilk3"),
    # made plot background same color as panel
    plot.background = element_rect(fill = "cornsilk3"))
```



e.

Figure 2

Temperature Influences my Distance Walked

The vertical position of the outlined, purple points represent the distance I walked in one day in kilometers, and the horizontal position of those points represents the high temperature in Celsius that day. The brown line represents the predicted distance I walked as a function of high temperature, which was generated using a linear model. The brown, shaded ribbon represents the 95% confidence interval from that linear model. Data was personally collected from 2026/04/23 to 2026/05/24. Distance walked was collected using the app `Map My Walk` and high temperature was collected using the app `Weather Underground`.

f.

I found a moderate, positive correlation (Spearman's $\rho = 0.415$) between the daily high temperature and the distance I walked that day. The high temperature accounted for 11.62% of the variation in distance walked (linear regression, F(1,30) = 5.075, $R^2 = 0.1162$, p = 0.03174, $\alpha = 0.05$), and with every one degree increase in temperature, the model predicted that I would walk an additional 0.3918 ± 0.1739 (SE) km.

Contrary to my initial prediction (of an inverse correlation between temperature and distance walked), warm weather actually increased the amount I walked, likely because cold weather caused me to hunker inside, and cold weather is associated with fog, which also causes me to avoid going outside. However, the low R^2 shows that other variables likely played a larger role in how much I walked each day.